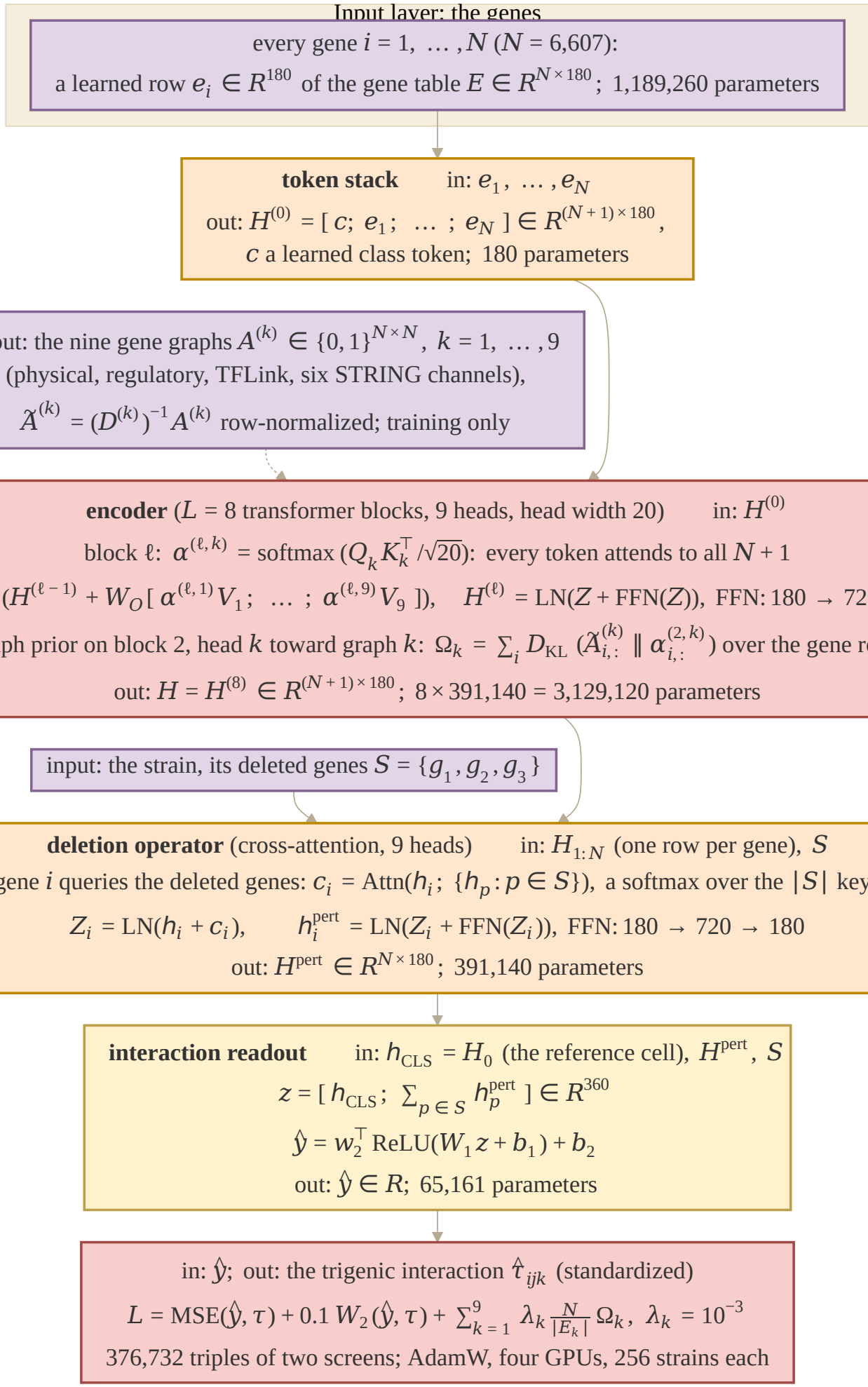

Input layer: the genes
every gene $i = 1, \ldots, N$ (N = 6,607):
a learned row $e_i \in R^{180}$ of the gene table $E \in R^{N \times 180}$; 1,189,260 parameters
token stack in: $e_1, \ldots, e_N$
out: $H^{(0)} = [\, c; e_1; \ldots; e_N \,] \in R^{(N+1) \times 180}$,
c a learned class token; 180 parameters
input: the nine gene graphs $A^{(k)} \in \{0,1\}^{N \times N}$, $k = 1, \ldots, 9$
(physical, regulatory, TFLink, six STRING channels),
$\tilde{A}^{(k)} = (D^{(k)})^{-1} A^{(k)}$ row-normalized; training only
encoder (L = 8 transformer blocks, 9 heads, head width 20) in: $H^{(0)}$
block ℓ: $\alpha^{(\ell,k)} = \mathrm{softmax}(Q_k K_k^{\top}/\sqrt{20})$: every token attends to all $N+1$
$Z = \mathrm{LN}(H^{(\ell-1)} + W_O[\, \alpha^{(\ell,1)} V_1; \ldots; \alpha^{(\ell,9)} V_9 \,])$, $H^{(\ell)} = \mathrm{LN}(Z + \mathrm{FFN}(Z))$, FFN: 180 → 720 → 180
graph prior on block 2, head k toward graph k: $\Omega_k = \sum_i D_{\mathrm{KL}}(\tilde{A}^{(k)}_{i,:} \,\|\, \alpha^{(2,k)}_{i,:})$ over the gene rows
out: $H = H^{(8)} \in R^{(N+1) \times 180}$; $8 \times 391{,}140 = 3{,}129{,}120$ parameters
input: the strain, its deleted genes $S = \{g_1, g_2, g_3\}$
deletion operator (cross-attention, 9 heads) in: $H_{1:N}$ (one row per gene), S
gene i queries the deleted genes: $c_i = \mathrm{Attn}(h_i; \{h_p : p \in S\})$, a softmax over the $|S|$ keys
$Z_i = \mathrm{LN}(h_i + c_i)$, $h_i^{\mathrm{pert}} = \mathrm{LN}(Z_i + \mathrm{FFN}(Z_i))$, FFN: 180 → 720 → 180
out: $H^{\mathrm{pert}} \in R^{N \times 180}$; 391,140 parameters
interaction readout in: $h_{\mathrm{CLS}} = H_0$ (the reference cell), H^{pert}, S
$z = [\, h_{\mathrm{CLS}}; \sum_{p \in S} h_p^{\mathrm{pert}} \,] \in R^{360}$
$\hat{y} = w_2^{\top} \mathrm{ReLU}(W_1 z + b_1) + b_2$
out: $\hat{y} \in R$; 65,161 parameters
in: $\hat{y}$; out: the trigenic interaction $\hat{\tau}_{ijk}$ (standardized)
$L = \mathrm{MSE}(\hat{y}, \tau) + 0.1\, W_2(\hat{y}, \tau) + \sum_{k=1}^{9} \lambda_k \frac{N}{|E_k|} \Omega_k$, $\lambda_k = 10^{-3}$
376,732 triples of two screens; AdamW, four GPUs, 256 strains each